architecture that uses EC2 isn't fault tolerant by default. But AWS provides the building blocks to help you improve the fault tolerance of virtual machines. In this chapter, we will show you how to use Auto Scaling groups, Elastic Load Balancing (ELB), and Simple Queue Service (SQS) to turn EC2 instances into fault-tolerant systems.

First, however, let's look at the level of failure resistance of key services. Knowing which services are fault tolerant, which are highly available, and which are neither will help you create the kind of fault tolerance your system needs.

The following services provided by AWS are neither highly available nor fault tolerant. When using one of these services in your architecture, you are adding a *single point of failure* (SPOF) to your infrastructure. In this case, to achieve fault tolerance, you need to plan and build for failure as discussed during the rest of the chapter:

- *Amazon Elastic Compute Cloud (EC2) instance*—A single EC2 instance can fail for many reasons: hardware failure, network problems, availability zone (AZ) outage, and so on. To achieve high availability or fault tolerance, use Auto Scaling groups to set up a fleet of EC2 instances that serve requests in a redundant way.
- *Amazon Relational Database Service (RDS) single instance*—A single RDS instance could fail for the same reasons that an EC2 instance might fail. Use Multi-AZ mode to achieve high availability.

All the following services are *highly available* (HA) by default. When a failure occurs, the services will suffer from a short downtime but will recover automatically: